

Portfast

- Directly Put the port in the forwarding state instead of waiting STP elections

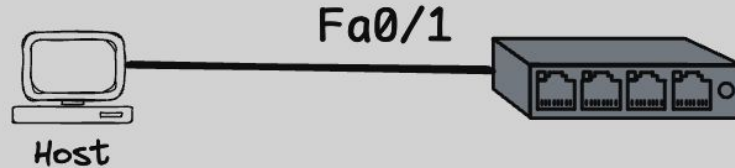
Discard

Learn

Forwarding

- Doesn't generate TCN when it goes down & is not impacted by any TCN in the network
- Configured on non switch ports

(Config-if)#spanning-tree portfast

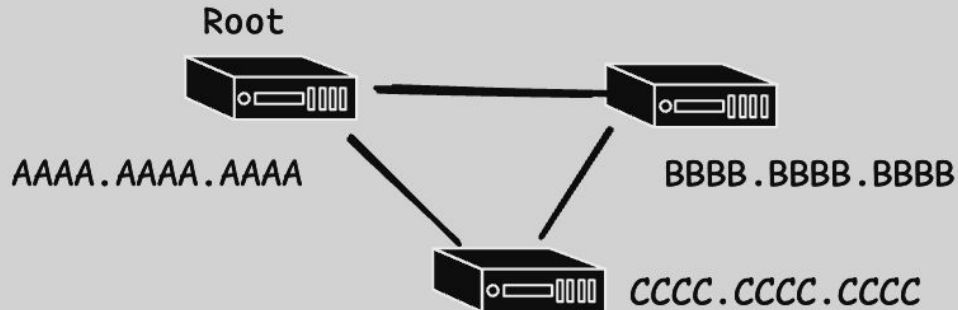


Root GUARD

-prevents unauthorized switches from becoming the Root Bridge in a network.

-If a better BPDU is received, the port is placed in root-inconsistent state (blocked)

```
(Config-if)# spanning-tree guard root
```



Err Disable

-Automatically by default disables a switch port when it detects certain network issues

Port Security Violation – Unauthorized MAC address detected.

Link Flapping – The port goes up and down frequently.

BPDU Guard – A BPDU is received on a PortFast-enabled port.

To Recover the port Manually

```
interface GigabitEthernet0/1  
shutdown  
no shutdown
```

Recover Automatically (not the default)

#errdisable recovery cause bpduguard

```
errdisable recovery cause all  
errdisable recovery interval 30
```

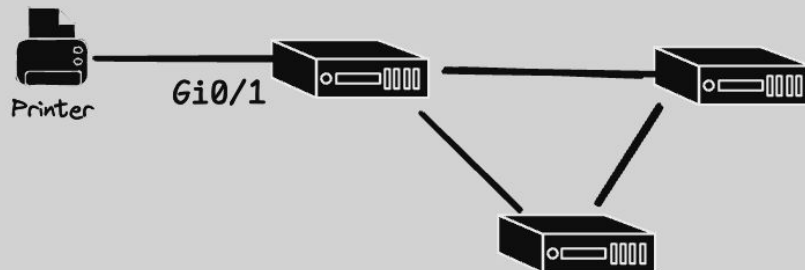
BPDUGUARD

-When Interface Receives BPDU it will go into Err disable mode

```
(config)#spanning-tree portfast bpduguard default
```

```
(Config)# Interface gi0/1
```

```
(Config-if)# Spanning-Tree BPDUGuard Enable
```



```
%SPANTREE-2-BLOCK_BPDUGUARD: Received BPDU on port GigabitEthernet0/1 with BPDU Guard enabled. Disabling port.
```

```
%PM-4-ERR_DISABLE: bpduguard error detected on Gi0/1, putting Gi0/1 in err-disable state
```

```
SW3# show interfaces g0/1
```

```
GigabitEthernet0/1 is down, line protocol is down (err-disabled)
```

BPD GUARD

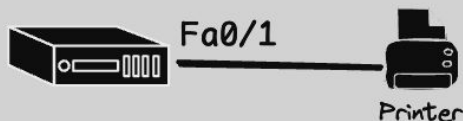
-Filter BPDU inbound & outbound

```
(config)#spanning-tree portfast bpduguard default
```

once Recieve bpdu, it will lose portfast and start negotiating STP

```
(Config)# Interface Fa0/1
```

```
(Config-if)# Spanning-Tree BPDUGuard Enable
```



Loop GUARD

-Enabled on Root/Block port

```
(config)#spanning-tree loopGuard default
```

```
(Config-if)# Spanning-Tree Guard loop
```

